

Validating Multi-Model LLM Ensembles for Survey Harmonization Classification

DRAFT — not validated — see disclaimer slide

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Disclaimer



Warning

DRAFT — These materials are in development and have not been independently validated.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

Research Question

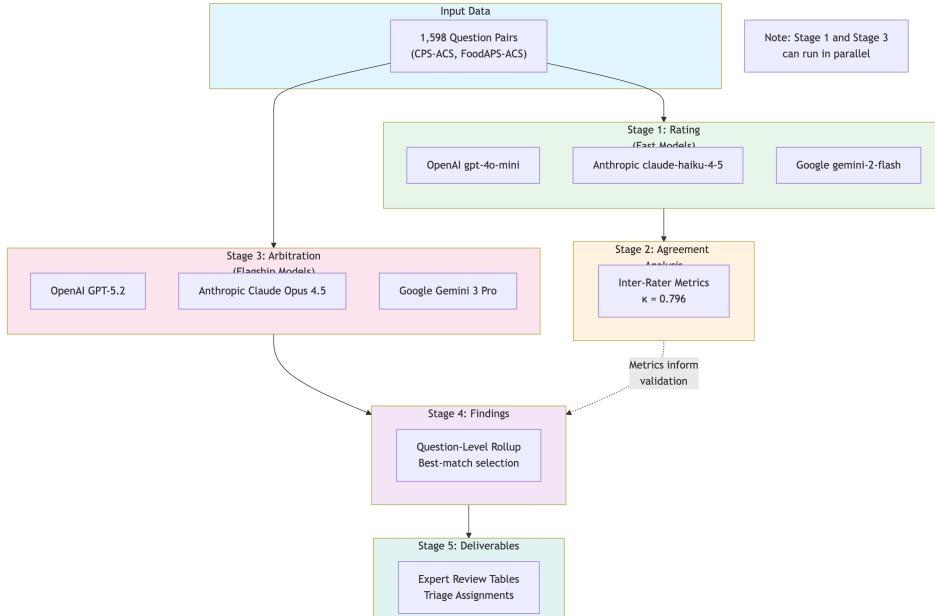
Can multi-model LLM ensemble reliably classify survey question harmonization compatibility?

- ▶ **Challenge:** Survey harmonization requires nuanced expert judgment
- ▶ **Approach:** Three-tier architecture (raters → arbitrators → triage)
- ▶ **Validation:** Construct validity through independent model convergence
- ▶ **Analysis:** Behavioral patterns in arbitrator decision-making

Pairwise Comparison Task

- ▶ Within concept categories, compare all question pairs
- ▶ **1,598 pairs** analyzed (CPS-ACS, FoodAPS-ACS)
- ▶ Harmonization classification: F1/F2/F3 with barrier codes
- ▶ Naive but exhaustive — expect many F3, looking for matches

Three-Tier Architecture



Multi-Model Ensemble Design

Three frontier models to reduce bias:

Model	Behavior
OpenAI	Moderate synthesis, slight pro-self bias
Anthropic	High synthesis, neutral
Google	Deferential, conservative

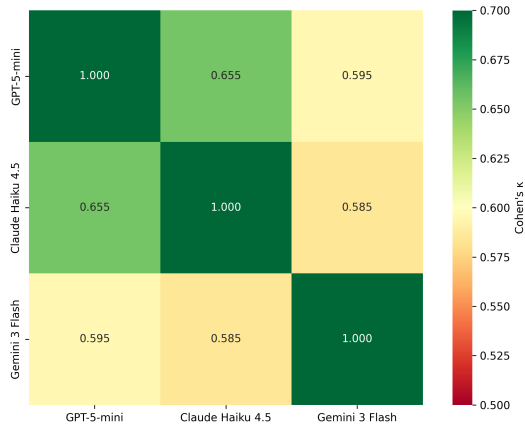
Different profiles = documented finding, not noise

Construct Validity: Why Trust These Results?

Three independent architectures → same judgments = well-defined task

Rater Agreement

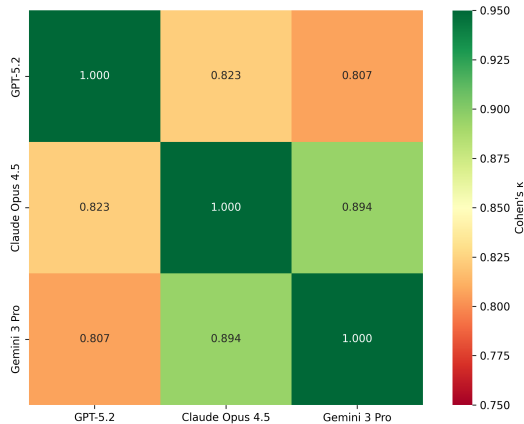
Rater Inter-Model Agreement (Cohen's κ)



Fleiss $\kappa = 0.611$ (substantial)

Arbitrator Agreement

Arbitrator Inter-Model Agreement (Cohen's κ)



Fleiss $\kappa = 0.843$ (almost perfect)

Rater Agreement

- ▶ **= 0.611** (Fleiss) — substantial agreement
- ▶ Three independent architectures converge on same patterns
- ▶ Disagreement rate manageable for arbitration tier

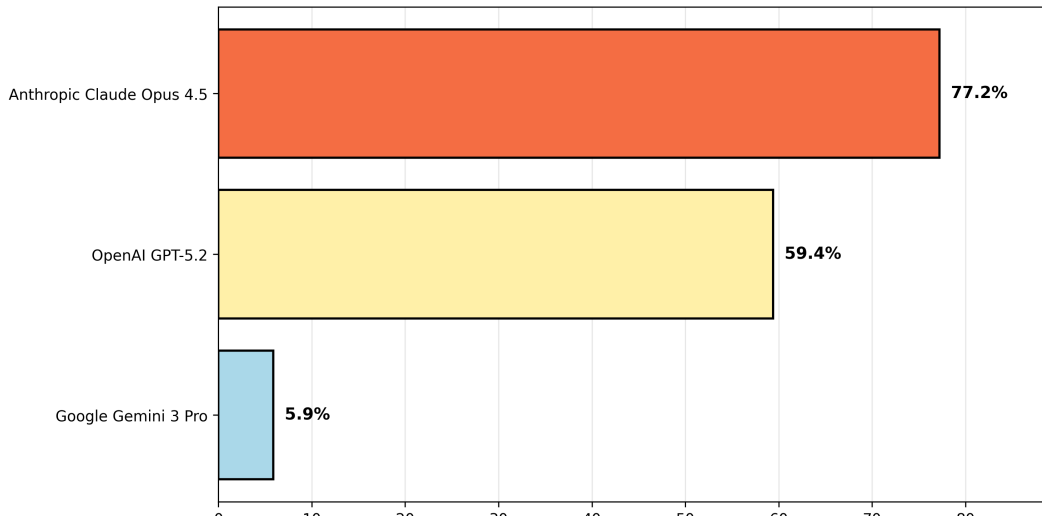
Arbitrator Agreement

- ▶ = **0.843** (Fleiss) — almost perfect agreement
- ▶ Flagship models with deeper reasoning → higher convergence
- ▶ Full arbitration coverage enables behavioral analysis across all pairs

Arbitrator Behavioral Differences

Not a bug — a feature. Diversity captures multiple valid interpretations.

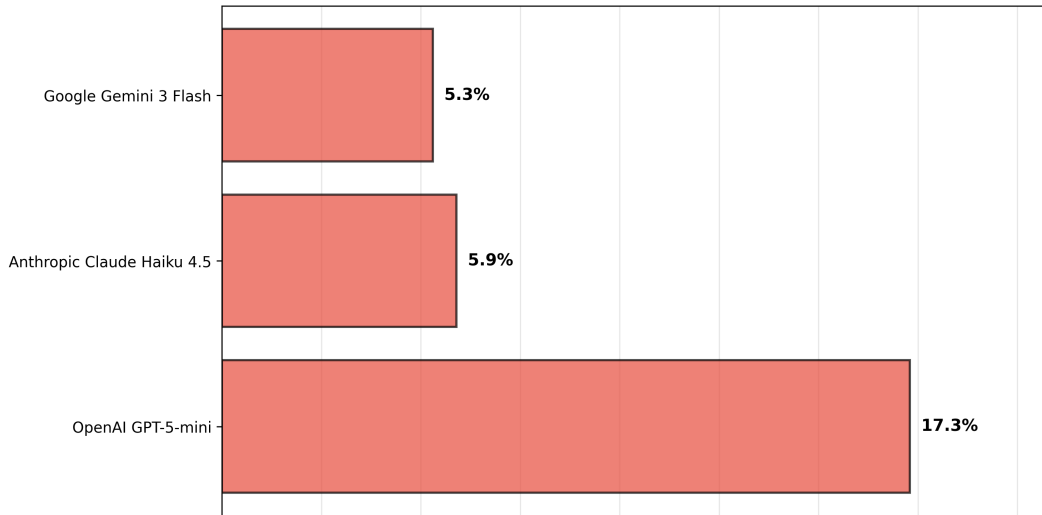
Arbitrator Synthesis Behavior



Why Not Just Use One Model?

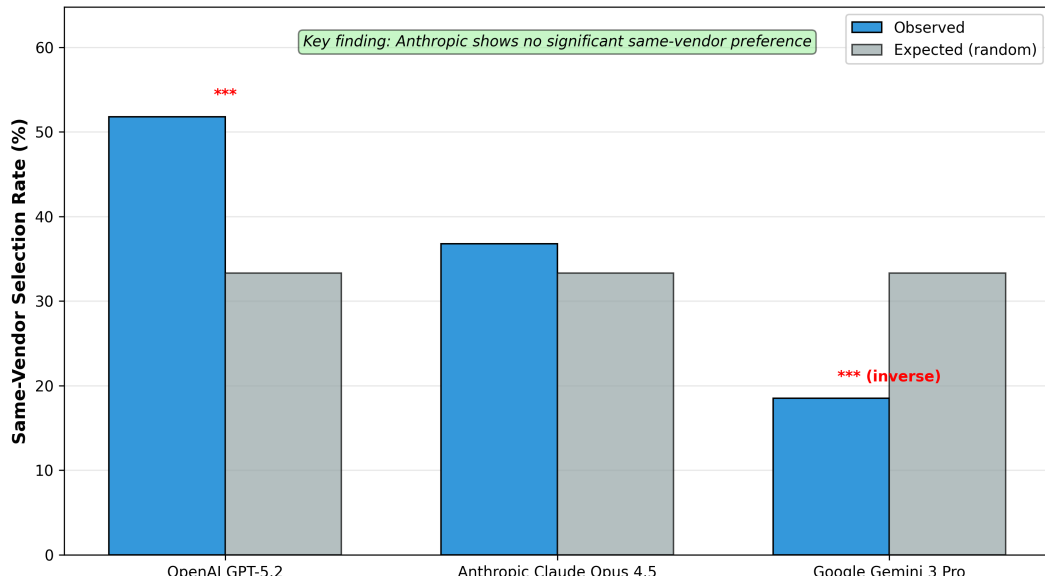
Single-model divergence from ensemble majority:

Single-Model Risk: Divergence from Ensemble



Do Arbitrators Favor Their Own Vendor's Rater?

Arbitrator Vendor Bias Analysis



Cost/Quality: Is the 3rd Model Worth It?

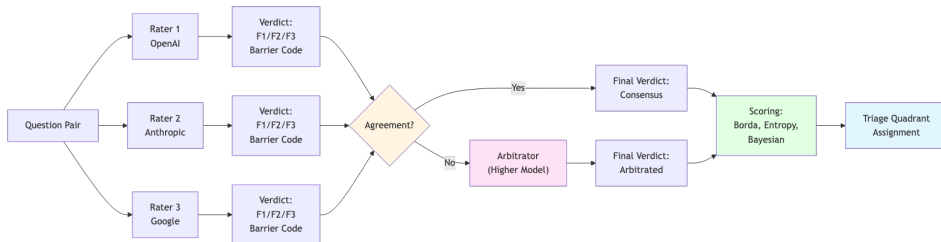
Two-model scenarios — tie frequency:

Pair	Feasibility Ties
OpenAI + Anthropic	23.2%
OpenAI + Google	22.1%
Anthropic + Google	11.0% ← lowest

Recommendation: - <1,000 pairs: 2 models + human tiebreaking - >1,000 pairs: 3 models for automated triage

The 3rd model's value is highest when first two disagree.

LLM Ensemble Pattern



Methodology Validation Summary

Validation Check	Result	Interpretation
Rater convergence	= 0.611	Substantial — task well-defined
Arbitrator convergence	= 0.843	Almost perfect — disagreements resolvable
Single-model risk	5-17%	Multi-model approach justified
Vendor bias	Mixed	Anthropic unbiased; OpenAI self-favoring

Conclusion: Multi-model ensemble provides robust classifications suitable for survey harmonization analysis.

Implications for AI-Assisted Research

- ▶ **Construct validity** through independent architecture convergence
- ▶ **Behavioral diversity** provides multiple valid perspectives
- ▶ **Cost/quality tradeoffs** enable scaling to larger datasets
- ▶ **Transparent biases** better than hidden single-model assumptions

Reusable methodology for future survey harmonization work

What's Next

1. Apply to additional federal surveys
2. Extend to survey editing/imputation use cases
3. Investigate optimal ensemble sizes for different task types

Summary

Research Question: Can multi-model LLM ensembles reliably classify survey harmonization?

Answer: Yes, with documented construct validity

Evidence: - High convergence ($\kappa = 0.611 \rightarrow 0.843$) - Documented behavioral differences - Quantified single-model risk - Vendor bias patterns identified

Impact: Validated methodology for AI-assisted survey analysis

Questions?

Contact: [your info]

Repository: [link]

Methodology documentation: [link]

Appendix

Harmonization Framework

Adopted standard survey harmonization classification:

Level	Definition
F1	Directly consolidable — no transformation needed
F2	Consolidable with transformation — needs recoding
F3	Not consolidable — fundamental barriers

Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

Harmonization Taxonomy

Primary Source: Fortier et al. (2011, 2017) DataSHaPER/Maelstrom framework

Supporting: Wolf et al. (2016), Saris & Gallhofer (2014), Slomczynski & Tomescu-Dubrow (2018)

Code	Feasibility	Definition	Action Required
F1	Direct recode	Mechanically transformable with simple recoding or collapsing	Simple data transformation
F2	Statistical adjustment	Requires modeling, imputation, or bridging studies	Statistical harmonization
F3	Incompatible	Fundamentally different constructs; not harmonizable without re-fielding	No harmonization possible

Barrier Code Taxonomy (1/3)

Temporal Constraints (TC)

Code	Subtype	Definition	Example
TC	Temporal/Chronological	Reference period or timing differences	—
TC.1	Reference period length	Different recall windows	7-day vs 12-month
TC.2	Temporal framing	Point-in-time vs habitual vs retrospective	“last week” vs “usually”
TC.3	Calendar alignment	Fixed vs rolling reference periods	Calendar year vs “past 12 months”

Construct Constraints (CC)

Code	Subtype	Definition	Example
CC	Construct / Content	Construct definition	

Barrier Code Taxonomy (2/3)

Population/Coverage Constraints (PC)

Code	Subtype	Definition	Example
PC	Population/Coverage	Universe, frame, or sample design differences	—
PC.1	Universe definition	Target population differs	Civilian vs all residents
PC.2	Frame exclusions	Different sampling exclusions	Group quarters, territories
PC.3	Age bounds	Different age eligibility	15+ vs 16+ vs 18+
PC.4	Geographic scope	Different geographic coverage	50 states vs states + territories

Response Scale Constraints (RS)

Code	Subtype	Definition	Example
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Barrier Code Taxonomy (3/3)

Mode/Context Constraints (MC)

Code	Subtype	Definition	Example
MC	Mode/Context	Interview mode or questionnaire context differences	—
MC.1	Interview mode	Different data collection modes	CATI vs web self-administered
MC.2	Question routing	Different skip patterns	Subset vs all respondents
MC.3	Contextual priming	Preceding questions affect interpretation	Question order effects
MC.4	Proxy response	Proxy vs self-report rules	Household head vs individual

Processing/Metadata Constraints (PM)

Key Citations

Harmonization Framework:

- ▶ Fortier, I., et al. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
- ▶ Fortier, I., et al. (2017). Maelstrom Research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, 46(1), 103-115.

Survey Methodology:

- ▶ Wolf, C., et al. (2016). Harmonizing survey questions between cultures and over time. In *The SAGE Handbook of Survey Methodology* (pp. 502-524). SAGE.
- ▶ Saris, W. E., & Gallhofer, I. N. (2014). *Design, Evaluation, and Analysis of Questionnaires for Survey Research* (2nd ed.). Wiley.

Data Harmonization:

- ▶ Slomczynski, K. M., & Tomescu-Dubrow, I. (2018). Basic principles of survey data recycling. In *Advances in Comparative Survey Methods* (pp. 937-962). Wiley.
- ▶ Tomescu-Dubrow, I., et al. (Eds.). (2024). *Survey Data Harmonization in the*